

Indus seal inscriptions as credentials, not names: calibrated and pre-registered tests, with the Umma sealing as an exported standard formula

Draft, 30 September 2026 (corrected: the Umma sealing is in the Ashmolean, not the Louvre). Authors: P. Garg (and co-authors to be agreed). The code and data for every number are in this repository; the strategy IDs (S###) point to the log in STRATEGIES.md.

Abstract

Most attempts to decipher the Indus script assume that seal inscriptions record owners' names, spelled phonetically. Mukhopadhyay (2023) argued instead that the inscriptions contain no proper nouns and served licensing, taxation and access control. We test this "no proper nouns" model quantitatively for the first time, on the merged Wells/ICIT corpus (3,599 distinct texts), with Mahadevan's 1977 transcription (IM77) as a transcription check. (1) **Calibrated name test.** Real seal-owner names from Mesopotamia (6,240 names, one legend per seal, CDLI) repeat far more than a bigram generator trained on them predicts: the uniqueness ratio is 0.62. The Indus seal 'middles' do not repeat any more than such a generator predicts: the ratio is 0.97. (2) **Pre-registered credential test.** We model a seal text as a credential: an introductory 'shield' sign, then holdings or dealings, then an office word. We froze two predictions before testing and checked them on 239 texts from sites absent from IM77. Short texts recur whole inside longer ones at $4.1\times$ the within-text-shuffled rate, and 40% of the held-out texts reuse a ≥ 3 -sign run known from Mohenjo-daro or Harappa ($5\times$). (3) **Nameless quantity seals.** Thirty-one seals from eight sites carry only 'shield · of · N · TREE', with $N = 3\text{--}6$, the same counts as the Harappa voucher tablets. The Indus sealing from Umma (southern Iraq) carries this same formula, '... of 4 TREE', which appears as the complete text of seven home seals. (4) **Seal-sealing flow.** All 18 clay sealings that match a seal text were found at a different site from that seal, mostly in the direction Mohenjo-daro → Lothal and northern towns. Together these results support reading Indus seal texts as standardised credentials rather than personal names. We also propose a testable anchor: the tree sign as a timber unit, since timber (gesh ab-ba Meluhha) is the Meluhhan good most often recorded at Umma.

1. Background

- **Structure.** Indus seal texts have fixed beginners and enders, a fixed direction and correlated sign order (Yadav et al. 2009; Rao et al. 2009). Long texts divide into recurring segments (Yadav, Vahia, Mahadevan & Joglekar, 'Segmentation of Indus texts').
- **Function.** Mukhopadhyay (2023, *Humanities and Social Sciences Communications*) argued, largely from archaeological context and qualitative script-internal evidence, that the inscriptions encode no proper nouns and served taxation, trade and craft licensing and access control. Earlier readings (Mahadevan, Parpola) took the texts to be names plus titles.
- **Decipherment by statistics.** Statistical decipherment can produce convincing but wrong readings: planted keys 'read' the same symbols as English, Sanskrit or Tamil (arXiv 2608.02999). Our own planted-language controls agree (S107–S137).
- **What is missing, and what this paper adds.** No study has tested the no-proper-noun claim against real names with a calibrated null model, or tested a structural model of the texts on held-out sites with pre-registered predictions. This paper does both.

2. Data

- **Main corpus.** The merged Wells/ICIT corpus (`data/derived/merged-corpus-canonical.json`): 5,369 inscribed objects, 3,599 distinct texts after deduplication by site and text, with object type, emblem, find-spot and dimensions (`data/raw/inscriptions.csv`).
- **Transcription check.** IM77 (Mahadevan 1977), used for transcription-robustness checks. About 70% of its texts are the same objects as in the main corpus (S312), so agreement with IM77 is not independent replication.
- **Real names.** Mesopotamian seal legends from CDLI (ATF export).
- **Meluhha attestations.** The CDLI Meluhha attestations (214 texts).
- **Held-out set.** Texts from sites absent from IM77 (239 texts; S349).

3. Results

3.1 The middles are not a stock of personal names (S321–S322)

Method. For each text we remove the frame: the opener with its connective, and the closer with its suffix. We then compare the share of unique 'middles' with a bigram model trained on the same middles. The ratio of observed uniqueness to model uniqueness is the statistic. We apply the same procedure to the Mesopotamian seal-owner names (line 1 of each legend, one legend per seal), matching the sample size (1,748) over 50 draws.

Results. - Real names: 42.5% unique, against 68.6% under the bigram model (ratio 0.62). Common names recur: ur-Baba 111 times, lu2-dingir-ra 78, ur-Lamma 76. - Indus middles: 84.6% unique, against 87.4% under the model (ratio 0.97).

A population's names repeat because people share names; the Indus middles do not repeat in this way. Caveat: the Mesopotamian names are written syllabically, so the unit differs, but the ratio is measured against each corpus's own null model.

3.2 A credential model, tested on held-out sites (S347–S349)

Model. A seal text consists of: - a shield or opener (leaf, diamond, wheel, bracket or X) plus a connective; - holdings or dealings (counts of goods; fish-class words); - an office word from a closed set of about ten mutually exclusive closers, each with its own qualifiers (S289–S303).

In-sample results. - Short texts recur inside longer ones 4× more than when signs are shuffled within each text: 15.2% against 3.7%. - The longer and shorter versions of a credential show no ordering in time ($p = 0.40$), so they are parallel grades rather than promotions.

Pre-registered test. Predictions were frozen in `prereg/preregistration-4-credential.json` before the test, and checked on the held-out sites.

Prediction	Held-out result	Shuffled control	Ratio
P1: nesting	13.6%	3.3%	4.1×
P2: reuse of big-city runs	40.2%	7.9%	5.1×

Both predictions pass.

3.3 Nameless quantity seals and the Umma sealing (S335–S346)

- **Quantity seals.** Thirty-one texts consist only of 'opener · 2 · N · item', mostly with the item W390 (the tree or branch sign). They come from 8 sites and include two Kalibangan sealings, so the seals were used. Within Mohenjo-daro they are not clustered in one quarter, and seal size does not rise with N ($\rho = -0.38$, $p = 0.14$).
- **Counts.** On seals, the numbers attached to the tree sign peak at 3 and 4. The Harappa voucher counts (tall 2–4 + W700) peak at the same values.
- **The Umma sealing.** The Umma sealing (Ashmolean Museum, Oxford, 1931.120; published by Scheil 1925; Parpola 1994 no. 25) reads 127-705 · 2 · 4 · 390. Its ending, '2-4-390', is the complete text of seven home seals (Mohenjo-daro M-103, M-278, M-984, M-1750, M-1844; Harappa H-2003 and H-55; Dholavira; Surkotada). We are not aware of this equivalence being noted before.
- **Candidate anchor.** In CDLI texts about Meluhha, the most frequently named good is a wood, {gesz}ab-ba me-luh-ha, used for furniture at Umma (P107404, P249043), Ur, Girsu and Isin. We therefore propose W390 = a timber unit. This is a hypothesis with a stated test: sealing backs (log or rope impressions), and Mesopotamian wood accounts with counts of 3–6 units. A point against it: the back of the Umma bulla carries the impression of textile, so it was attached to a cloth-covered bale (Parpola 1994), which fits bundled goods better than logs. The find-spot 'Umma' also rests only on the dealer's word (Scheil 1925).

3.4 Where sealed goods went (S324–S330)

- **All matches are cross-site.** Eighteen sealings match a seal text exactly, against about 1 expected by chance. All 18 were found at another site: Mohenjo-daro → Lothal 13, and → Harappa, Kalibangan, Hulas and Rupar; one match runs Harappa → Lothal.
- **Lothal.** The ten Lothal bale sealings (L-161–170) carry 817-2-48-740 with an **elephant**. The only seal with this text, Mohenjo-daro M-895, has a **unicorn**, and Lothal has no elephant seals at all (0 of 43). So the same credential was held under different emblems, and the bales were sealed away from Lothal.
- **Kish.** A second Kish seal reproduces the full text of Mohenjo-daro seal M-94 (3-220-590-390-740), preceded by a unit, 416-840, that also occurs at Mohenjo-daro, Dholavira and Harappa. An identical Kish–Mohenjo-daro pair (Mackay 1925 / M-228) has been noted before; this is a second such case.
- **Harappa tablets.** Harappa tablets reproduce seal texts from Mohenjo-daro, Lothal and Dholavira, e.g. 415-220-520, which occurs on 15 Harappa tablets and 1 Mohenjo-daro seal (S344).

4. Discussion

The results agree with Mukhopadhyay's (2023) thesis and add calibrated, pre-registered and held-out tests. The emblem is independent of the text: emblem × title association $P = 0.38$ (S329); the same text appears under different animals (S325). The script behaves as a word-sign labelling system with a fixed slot grammar, not as recorded speech: repetition is formula-like, there are no spelling traces, and phonetic keys fail their controls. On this view, 'decipherment' means identifying what the credential terms referred to (goods, counts, offices) through outside matches, not recovering a spoken language.

5. Limitations

- The corpus is a small sample of all seals ever made: Chao1 estimates about 950 signs, of which about 240 are unseen (S309). Most texts come from Mohenjo-daro and Harappa (76%).
- Sign identities depend on the transcription. Allograph merges were run at three levels (strict, strong, all), and the results hold at each.

- The Mesopotamian comparison mixes a syllabic script with a word-sign script. The within-corpus null model mitigates this, but does not remove it.
- The timber reading of W390 is a hypothesis, not a demonstration, and the textile impression on the back of the Umma bulla argues against it. The Umma provenance is a dealer's report.
- A literature check (web; Mukhopadhyay 2023; Yadav et al.; Laursen 2010; arXiv 2608.02999) found no prior report of results 3.1, 3.2 (pre-registered form), 3.3 or 3.4 (cross-site statistics). A specialist check of Parpola's CISI commentary is still pending.

6. Reproducibility

The scripts are in `tools/`. Each is named after its STRATEGIES row: - `strat_namecalib.py`, `strat_namecalib2.py` (S321–S322) - `strat_credential.py`, `strat_credential_heldout.py` (S347, S349) - `strat_batch1.py` (S336–S338) - `strat_sealflow.py` (S324) - `strat_branches.py`, `strat_590title.py`, `strat_abroadsuffix.py` (S326–S328) - `parse_all.py` (S310)

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